

Probability and Distributions

STA257H1 - Probability and Statistics I

Omidali A. Jazi

University of Toronto
Department of Statistical Sciences

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Goals for this Chapter

- Introduction
- Sets
- The Probability Set Function
- Conditional Probability and Independence
- Random Variables
- Discrete Random Variables
- Continuous Random Variables
- Expectation of a Random Variable
- Some Special Expectations
- Important Inequalities

Section 1

Introduction

Defining Probability

Example, 1.3.3

- In every day conversation, probability is a measure of a person's belief in the occurrence of a future event.

- **Example:** What is the chance of winning an election?

In this sense, probability is completely **subjective**. *→ based on intuition
personal opinion*

- More quantitatively, we define probability as the fraction of times an event occurs if it's repeated over and over (infinitely).

This is the **relative frequency** definition of probability.

*3. Axioms of
probability (later)*

*↳ repeat an experiment. Find the fraction of
times we observe an outcome.*

Probability vs Statistics

- What is the difference between probability and statistics?
- In probability, the **randomization mechanism** is assumed/known and calculations follow.

Example: Given a fair coin, what's the chance of observing 8 heads out of 10 coin flips?

- In statistics, the outcome is known and used to infer something about the mechanism.

Example: Given that I have just observed 8 heads out of 10 coin flips, is the coin fair?

test the claim

Probability Quantifies Outcome Uncertainty

- Imagine someone is flipping a coin. With enough practice, we can learn to flip a coin and have it come up heads every time.

In this case, there is no uncertainty in the outcome. But without practice, the coin is going to come up heads and tails "randomly".

- There is randomness because the particular outcomes are uncontrollable and/or unpredictable.
- Probability helps you quantify the likelihood of various outcomes.

the randomness in experiments

Probability and Inference

- In statistics, a sample of data is used to say something about the entire population. *generalization*
- We learn formal methods for making inference such as hypothesis testing and model fitting. *STA261*
- But first, we need to understand the theory of probability. Both what it is and, perhaps more importantly, how to calculate it.

Section 2

Sets

Terminologies and Notation

To begin, we need some terminologies and notation from the set theory.

sets: A collection of objects or people

- Capital letters denote the sets of objects: A , B , etc.
 *$\neq \{a_1, a_1, a_2, a_3\}$
 $A = \{a_1, a_2, a_3\}$
 $\neq \{a_2, a_1, a_3\}$*
- If set A consists of objects a_1, a_2, a_3 , we write $A = \{a_1, a_2, a_3\}$
 $= \{a_1, a_1, a_2, a_3\} \neq \{a_2, a_1, a_3\}$
- S denotes the **universal set** which is the set of all possible objects.
- ϕ denotes the **null** or **empty set** ($\phi = \{\}$) which is the set without any object.

*phi
greek
letter*

Terminologies and Notation

$$A \subsetneq B \quad C$$

$$A \subseteq B$$

- For any two sets A and B , A is a **subset** of B if every object in A is in B . Notation: $A \subseteq B$.

includes the case $A = B$

- Example:** If $B = \{a_1, a_2, a_3, a_4\}$ and $A = \{a_1, a_2, a_3\}$ then A is a subset of B . $A \subset B$

$$A \subset S$$

- Note: The null set is a subset of every set. $\phi \subset A$

- The **union** of A and B is the set of all objects in A or B or both. Notation: $A \cup B$. *at least one set.*

- Example:** If $A = \{a_1, a_2\}$ and $B = \{b_1, b_2\}$ then $A \cup B = \{a_1, a_2, b_1, b_2\}$.

XOR (exclusive OR) = either one, or the other. But not both (exactly one)

Terminologies and Notation

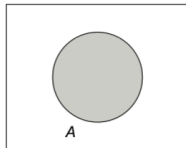
$$A \subset B \Rightarrow \begin{aligned} A \cap B &= A \\ A \cup B &= B \end{aligned}$$

- The **intersection** of sets A and B is the set of all objects in both A and B . Notation: $A \cap B$ or just AB .
- **Example:** If $B = \{a_1, a_2, a_3, a_4\}$ and $A = \{a_1, a_2, a_3\}$ then $A \cap B = \{a_1, a_2, a_3\}$.
- The key word for union is or while the key word for intersection is and.

Venn Diagrams

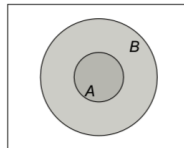
universal set
↖ S

Panel (a)

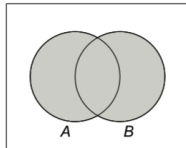


A

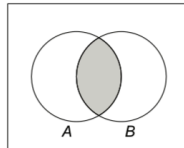
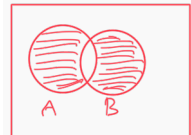
Panel (b)

 $A \subset B$

Panel (c)

 $A \cup B$

Panel (d)

 $A \cap B$  $A \oplus B$

XOR

Terminologies and Notation

$$A', \overline{A}$$

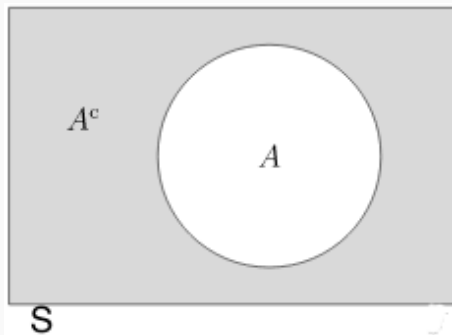
- If A is a subset of S , then the **complement** of A is denoted by A^c .
- The complement of A is the set of all objects in S that are not in A .
- Note that $A \cup A^c = S$, $A \cap A^c = \phi$
- Two sets A and B are said to be **disjoint** or **mutually exclusive** if they have no object in common. That is, A and B are disjoint if $A \cap B = \phi$

Note: By definition, $A \cap A^c = \phi$. A and A^c are disjoint (mutually exclusive)

$$\phi \subset A$$

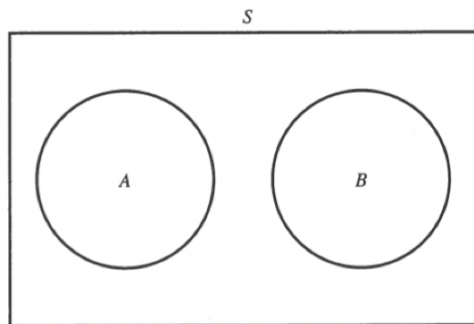
$$\phi \cap A = \phi$$

Venn Diagram: Complement



Venn Diagram: Mutually Exclusive

Venn diagram for
mutually exclusive
sets A and B



Laws

$$A \cup (B \cap C) = (A \cup B) \cap C \Rightarrow \text{Associate law}$$

$$A \cap (B \cup C) = (A \cap B) \cup C$$

- Distributive Laws:

$$\underline{A} \cap (\underline{B} \cup C) = (\underline{A} \cap B) \cup (\underline{A} \cap C)$$

$$A \cup (\underline{B} \cap C) = (\underline{A} \cup B) \cap (\underline{A} \cup C)$$

$2 \times (3+4) = 2 \times 3 + 2 \times 4$
 $24 = 2 + (3 \times 4) \neq (2+3) \times (2+4) = 30$ Extension: $A \cap (\bigcup_{i=1}^n B_i) = \bigcup_{i=1}^n (A \cap B_i)$
 • DeMorgan's Laws: $A \cup (\bigcap_{i=1}^n B_i) = \bigcap_{i=1}^n (A \cup B_i)$

$$(A \cap B)^c = A^c \cup B^c$$

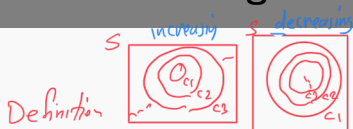
$$(A \cup B)^c = A^c \cap B^c$$

$$A_1 \cup A_2 \cup \dots = \bigcup_{n=1}^{\infty} A_n$$

$$A_1 \cap A_2 \cap \dots = \bigcap_{n=1}^{\infty} A_n$$

$$\left(\bigcup_{n=1}^{\infty} A_n \right)^c = \bigcap_{n=1}^{\infty} A_n^c, \quad \left(\bigcap_{n=1}^{\infty} A_n \right)^c = \bigcup_{n=1}^{\infty} A_n^c$$

Nondecreasing & Nonincreasing Sets



Definition

$$C_1 = (0, 0) = \{\}$$

$$C_2 = (0, \frac{1}{2})$$

$$C_3 = (0, \frac{2}{3}), \dots$$

$$C_n = (0, 1 - \frac{1}{n}), n = 1, 2, 3, \dots$$

A sequence of sets $\{C_n\}$ is nondecreasing if $C_n \subset C_{n+1}$, for all n , in which we write

$$\lim_{n \rightarrow \infty} C_n = \bigcup_{n=1}^{\infty} C_n.$$

$$C_1 = \bigcup_{n=1}^{\infty} C_n$$

increasing or identical

A sequence of sets $\{C_n\}$ is nonincreasing if $C_{n+1} \subset C_n$, for all n , in which we write

$$\lim_{n \rightarrow \infty} C_n = \bigcap_{n=1}^{\infty} C_n.$$

$$C_1 = \bigcap_{n=1}^{\infty} C_n$$

decreasing or identical

$$C_n = (0, \frac{1}{n})$$

$$C_1 = (0, 1)$$

$$C_2 = (0, \frac{1}{2})$$

$$C_3 = (0, \frac{1}{3}), \dots$$

Example 1: Show that the sequence of sets $C_k = \{x : \frac{1}{k} \leq x \leq 3 - \frac{1}{k}\}$ for $k = 1, 2, 3, \dots$ is

increasing and find $\lim_{k \rightarrow \infty} C_k$.

$$\lim_{k \rightarrow \infty} C_k = \bigcup_{k=1}^{\infty} C_k = (0, 3), \quad \bigcap_{k=1}^{\infty} C_k = C_1 = [1, 2] \quad \left| \begin{array}{l} C_1 = [1, 2], \quad C_2 = [\frac{1}{2}, 2.5], \dots \\ \frac{1}{k+1} < \frac{1}{k} \\ 3 - \frac{1}{k+1} < 3 - \frac{1}{k} \end{array} \right.$$

Example 2: Show that the sequence of sets $C_k = \{x : 2 < x \leq 2 + \frac{1}{k}\}$ for $k = 1, 2, 3, \dots$ is

decreasing and find $\lim_{k \rightarrow \infty} C_k$.

$$\lim_{k \rightarrow \infty} C_k = \bigcap_{k=1}^{\infty} C_k = \emptyset$$

$$\bigcup_{k=1}^{\infty} C_k = C_1 = (2, 3]$$

$$2 + \frac{1}{k+1} < 2 + \frac{1}{k}$$

$$C_1 = (2, 3]$$

$$C_2 = (2, 2.5], \quad C_3 = (2, 2.33), \dots$$

Examples 1.2.3, 1.2.4

$$C_n = [2, 2 + \frac{1}{n}]$$

$$C_1 = [2, 3]$$

$$C_2 = [2, 2.5]$$

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Section 3

The Probability Set Function

Random Experiment, Event, Sample Space

Definition 1

A **random experiment** is an act or process of observation that leads to a single outcome which cannot be predicted with certainty.

Definition 2

The **sample space** of an experiment is the set of all possible outcomes (sample points) and is denoted by S .

↓
element, outcome

Definition 3

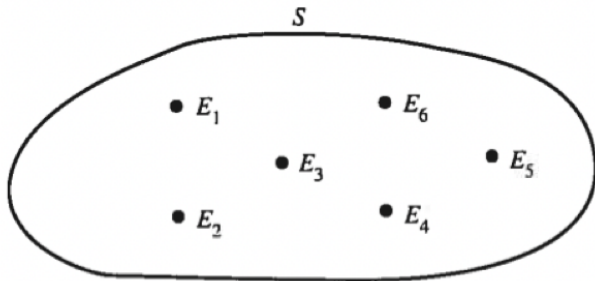
A **simple event** is an event that cannot be decomposed. Each simple event corresponds to one and only one **sample point** which is the most basic outcome of a random experiment. A **compound event** consists of two or more simple events.

Example

- In the six-sided die tossing experiment,
 $S = \{E_1, E_2, E_3, E_4, E_5, E_6\}$ where E_i represents "observe number i ".
↳ outcome (sample point) $A = \{E_1\} \Rightarrow$ simple event
- The letter E with a subscript will be used to denote a ~~simple event~~
 or the corresponding sample point. *sample point*
- The events "observe a 1," "observe a 3" and "observe a 5" are all
 simple events since they cannot be decomposed any further. Each
 simple event corresponds to only one sample point.
 $\{E_1\}$ $\{E_3\}$ $\{E_5\}$ (outcome)
- However, the event "observe an odd outcome" on the six-sided die
 experiment is a compound event since it is composed of three simple
 events: "observe a 1," "observe a 3," and "observe a 5".
 $\rightarrow A = \{E_1 \cup E_3 \cup E_5\} = \{E_1, E_3, E_5\}$

Example

Venn diagram for the
sample space
associated with
the die-tossing
experiment

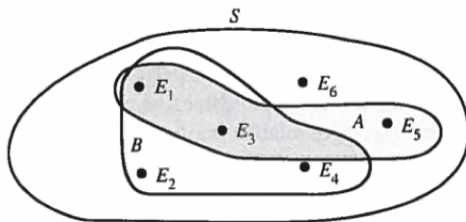


Example

The compound event A “observe an odd number” is $A = \{E_1, E_3, E_5\}$.

We could also write $A = \{E_1\} \cup \{E_3\} \cup \{E_5\}$. Define event B to be “observe a number less than 5” or $B = \{E_1, E_2, E_3, E_4\}$. $= \{E_1\} \cup \{E_2\} \cup \{E_3\} \cup \{E_4\}$

Venn diagram for the
die-tossing
experiment



Discrete Sample Space & Event

For uncountable sample space (S) , S is not rich enough to be closed under union and complement. That's why we need sigma field (this is beyond the scope of the course)

Definition 4

A **discrete sample space** is one that contains countable (finite or infinite) number of distinct sample points.

Definition 5

An **event** in a discrete sample space S is a collection of sample points. That is, any subset of S .

Example: A random experiment that involves tossing three dice at the same time until the same numbers come up.

$$6^3 = 216$$

↳ sample space is countable, infinite

Probability Axioms

On analyzing the relative frequency concept of probability, we see that three conditions must hold.

- ① The relative frequency of any event must be greater than or equal to zero. Hence, negative frequency does not make sense.
- ② The relative frequency of the whole sample space S must be 1 because all possible experimental outcomes are in S , something in S must occur every time the experiment is conducted.
- ③ If two events are mutually exclusive, then the probability of their union is the sum of their respective frequencies.

Probability Axioms

Definition 6

Suppose S is a sample space. To every event A in S , we assign a number $P(A)$, called the probability of A , so that the following axioms hold:

- Axiom 1: $P(A) \geq 0$
- Axiom 2: $P(S) = 1$
- Axiom 3: If $A_1, A_2, A_3, \dots$ are a sequence of **pairwise mutually exclusive** events in S (i.e, $A_i \cap A_j = \phi$ if $i \neq j$)

$$P(A_1 \cup A_2 \cup \dots) = P(A_1) + P(A_2) + \dots$$

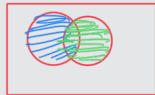
$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} P(A_n)$$

The Addition and Complement Rules

Theorem 7

(*The Addition Law of Probability*): The probability of the union of two events A and B is

$$✓ \quad P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



If A and B are mutually exclusive, then

$$✓ \quad \underbrace{A \cap B = \emptyset}_{P(\emptyset) = 0} \quad P(A \cup B) = P(A) + P(B) \quad ✓$$

Theorem 8

For an event A ,

$$P(A) = 1 - P(A^c).$$

Proof: In class.

Example 1: Gene Expression Profiling

Gene expression profiling is a state-of-the-art method for determining the biology of cells. In Briefings in Functional Genomics and Proteomics (Dec. 2006), biologists at Pacific Northwest National Laboratory reviewed several gene expression profiling methods. The biologists applied two of the methods (A and B) to data collected on proteins in human mammary cells.

The probability that the protein is cross-referenced (i.e., identified) by method A is 0.41, the probability that the protein is cross-referenced by method B is 0.42, and the probability that the protein is cross-referenced by both methods is 0.40.

$$P(A) = 0.41, \quad P(A \cap B) = 0.40$$

$$P(B) = 0.42$$

$$A \cap B \subset A \Rightarrow P(A \cap B) \leq P(A)$$

$$A \cap B \subset B \Rightarrow P(A \cap B) \leq P(B)$$

$$P(A \cap B) \leq \min\{P(A), P(B)\}$$

Example 1: Gene Expression Profiling

Example: Find the probability that the protein is identified by exactly one method.

$$P(A \oplus B) = P(A) + P(B) - 2P(A \cap B) = 0.41 + 0.42 - 2 \times 0.4 = 0.03$$



- Find the probability that the protein is cross-referenced by either method A or method B . ($A \cup B$ at least one method)

Solution: $0.41 + 0.42 - 0.40 = 0.43$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.41 + 0.42 - 0.4 = 0.43$$

Example: Find the probability that it's identified by A , but not B .

$$P(A \cap B^c) = P(A) - P(A \cap B) = 0.41 - 0.4 = 0.01$$



- On the basis of your answer to part a, find the probability that the protein is not cross-referenced by either method.

Solution: $1 - 0.43 = 0.57$

$$P(A^c \cap B^c) = P((A \cup B)^c) = 1 - P(A \cup B) = 1 - 0.43 = 0.57$$

{Calculating Probabilities: Sample-Point Method}

- ❶ First, define the experiment and clearly determine how to describe one simple event.
- ❷ List the simple events and test to make sure they cannot be decomposed. This defines S .
- ❸ Assign reasonable probabilities to the sample points in S making sure $P(A_i) \geq 0$ and $P(\cup_i A_i) = 1$. *Axiom 1,3*
- ❹ Define the event of interest A as a specific collection of sample points.
- ❺ Find $P(A)$ by summing the probabilities of the sample points in A .

Example

Consider a fair six-sided die. Let A_i ($i = 1, 2, \dots, 6$) denote the ^{simple} event “observe number i ”. Then, by Axioms 2 and 3:

$$\begin{aligned} P(S) &= P(A_1 \cup A_2 \cup A_3 \cup A_4 \cup A_5 \cup A_6) \\ &= P(A_1) + P(A_2) + P(A_3) + P(A_4) + P(A_5) + P(A_6) \\ &= 1 \end{aligned}$$

Counting Sample Points

- Let's look at some useful and powerful ways to count numbers of events.
- A sample space contains N equally likely events. An event A contains exactly n_a sample points, then we have

$$P(A) = \frac{n_a}{N} \begin{array}{l} \rightarrow \text{\# of sample points in } A \\ \rightarrow \text{\# of sample points in } S \end{array}$$

- So, if we know N and can count n_a , then we can calculate the probability of event A .

Pros & Cons of Sample-Point Method

- Sample-point approach is direct and powerful, but prone to human error.
- Mistakes include mis-defining simple events and missing simple events. For sample, spaces with large number of events, can be challenging (perhaps impossible) to count all the events.
- This leads us to learn about combinatorial analysis (combinations and permutations).

The mn Rule and Extension

Theorem 9

With m elements $\{a_1, a_2, \dots, a_m\}$ and n elements $\{b_1, b_2, \dots, b_n\}$, it is possible to form $m \times n$ pairs containing one element from each group.

Extension: If we have k sets of elements, n_1 in the first set, n_2 in the second set, ..., and n_k in the k th set. Suppose you wish to form a sample of k elements by taking one element from each of the k sets. Then the number of different samples that can be formed is $n_1 n_2 \dots n_k$.

The rule is also known as **Multiplication Rule**.

Permutations (Counting Ordered Arrangements)

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Definition 10

An ordered arrangement of r distinct objects is called a **permutation**. The number of ways of ordering n distinct objects taken r at a time is denoted by P_r^n .

without replacement

Theorem 11

To calculate the number of ways r items can be ordered out of n

$$P_r^n = n(n-1)(n-2) \cdots (n-r+1) = \frac{n!}{(n-r)!}$$

$$\binom{5}{2} = \frac{5 \times 4}{2} = 10$$

$$P(A) = \frac{\binom{2}{2} \binom{3}{0}}{\binom{5}{2}} = \frac{1}{10} \quad \Bigg/ \quad \frac{2}{5} \times \frac{1}{4} = \frac{1}{10}$$

Combinations (Counting Unordered Arrangements)

Definition 12

The number of **unordered** subsets of r objects out of n is called the number of **combinations**. That is, it is the number of subsets, of size r , that can be formed from n objects. The notation is C_r^n .

Theorem 13

The number of unordered subsets of size r chosen (without replacement) from n objects is

$$C_r^n = \binom{n}{r} = \frac{P_r^n}{r!} = \frac{n!}{r!(n-r)!}$$

The term $\binom{n}{r}$ is generally referred as the binomial coefficient because they occur in the binomial expansion $(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$

Combinations: Extension

Theorem 14

The number of ways of partitioning n distinct objects into k distinct groups containing $n_1, n_2, \dots, n_k$ objects, respectively, where each object appears in exactly one group and $\sum_{i=1}^k n_i = n$, is *(w, that's replacement, order is not important)*

$$\binom{n}{n_1, n_2, \dots, n_k} = \frac{n!}{n_1! n_2! \dots n_k!}$$

The terms $\binom{n}{n_1, n_2, \dots, n_k}$ are called the **multinomial coefficients** because they occur in the expansion of the multinomial term $(y_1 + y_2 + \dots + y_k)^n$.

Examples

- ✓ ● We have $n = 3$ people (A , B , and C) and $r = 2$ job positions. How many ways (permutations) are there to assign the positions to the people?

Solution: There are 6 orderings: $\{A, B\}$, $\{A, C\}$, $\{B, A\}$, $\{B, C\}$, $\{C, A\}$, $\{C, B\}$

- ✓ ● We have $n = 3$ people (A , B , and C) and pick $r = 2$ people to create a committee. How many combination of two types of three? (*Combination*)

Solution: There are only three: $\{A, B\}$, $\{A, C\}$, $\{B, C\}$.

- ✓ ● You need to assign 9 aircrafts to 3 sorties of size 4, 3, and 2 aircrafts. How many ways can the 9 aircrafts be assigned to the 3 groups?

Solution: $\frac{9!}{4!3!2!}$

*without replacement,
order is not important*

$$\binom{9}{4, 3, 2} = \frac{9!}{4!3!2!} = \frac{9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{4 \times 3 \times 2 \times 1 \times 3 \times 2 \times 1 \times 2 \times 1} = 9 \times 4 \times 7 \times 5$$

$$\begin{aligned} n_1 &= 4 \\ n_2 &= 3 \\ n_3 &= 2 \end{aligned} \quad \sum_{i=1}^3 n_i = 9$$

Additional Properties of Probability (1.3, 2)

Inclusion-Exclusion Formula: The Addition Law of Probability can be extended to k events, $A_1, A_2, \dots, A_k$ as

$$P\left(\bigcup_{i=1}^k A_i\right) = \sum_{i=1}^k P(A_i) - \sum_{i < j} P(A_i \cap A_j) + \dots + (-1)^{k-1} P(A_1 \cap \dots \cap A_k)$$

$$\begin{array}{l} k=1 \\ k=2 \\ \boxed{k=n} \checkmark \\ k=n+1 \end{array}$$

$$P\left(\bigcup_{i=1}^{n+1} A_i\right) = P\left(\left(\bigcup_{i=1}^n A_i\right) \cup A_{n+1}\right)$$

Proof: Exercise! (use induction)

$$P(A_1 \cup A_2 \cup A_3) = P(A_1) + P(A_2) + P(A_3) - P(A_1 \cap A_2) - P(A_1 \cap A_3) - P(A_2 \cap A_3) + P(A_1 \cap A_2 \cap A_3)$$

Additional Properties of Probability

Theorem 15

Let $\{C_n\}$ be an increasing sequence of events. Then

$$\lim_{n \rightarrow \infty} P(C_n) = P\left(\lim_{n \rightarrow \infty} C_n\right) = P\left(\bigcup_{n=1}^{\infty} C_n\right)$$

Let $\{C_n\}$ be a decreasing sequence of events. Then

$$\lim_{n \rightarrow \infty} P(C_n) = P\left(\lim_{n \rightarrow \infty} C_n\right) = P\left(\bigcap_{n=1}^{\infty} C_n\right)$$

$$P(A^c) = 1 - P(A)$$

Proof: In class.

$\{C_n^c\}$ are increasing

$$P\left(\bigcap_{n=1}^{\infty} C_n\right) = 1 - P\left(\left(\bigcap_{n=1}^{\infty} C_n\right)^c\right) = 1 - P\left(\bigcup_{n=1}^{\infty} C_n^c\right) = 1 - \lim_{n \rightarrow \infty} P(C_n^c)$$

This theorem is referred to as the **Continuity Theorem of Probability**.

Additional Properties of Probability

Another useful result for arbitrary unions is given by

Theorem 16

(Boole's Inequality) Let $\{C_n\}$ be an arbitrary sequence of events. Then

$$P\left(\bigcup_{n=1}^{\infty} C_n\right) \leq \sum_{n=1}^{\infty} P(C_n)$$

Proof: In class.

At least
one event occur

Hint: Consider $D_1 = C_1$ and $D_n = C_n - \bigcup_{i=1}^{n-1} C_i$ for $n = 2, 3, \dots$. Then

$\{D_n\}$ is a disjoint sequence of events, and further $\bigcup_{n=1}^{\infty} C_n = \bigcup_{n=1}^{\infty} D_n$.

Now, use Axiom 3 to prove the theorem.

Additional Properties of Probability

From the Addition Law of Probability, we have

$$P(C_1 \cup C_2) = P(C_1) + P(C_2) - P(C_1 \cap C_2) \leq 1$$

which gives **Bonferroni's Inequality** as

$$P(C_1 \cap C_2) \geq \underline{P(C_1) + P(C_2) - 1}$$
